EXPERIMENT – 6

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Design a Civic Issue Reporting System with AI-based (RAG-inspired) features to provide smart insights, classification, and status prediction.

AIM:

To design and simulate a Civic Issue Reporting application that incorporates basic concepts of Retrieval-Augmented Generation (RAG) using UI elements, providing intelligent feedback such as issue classification, status prediction, and contextual notifications.

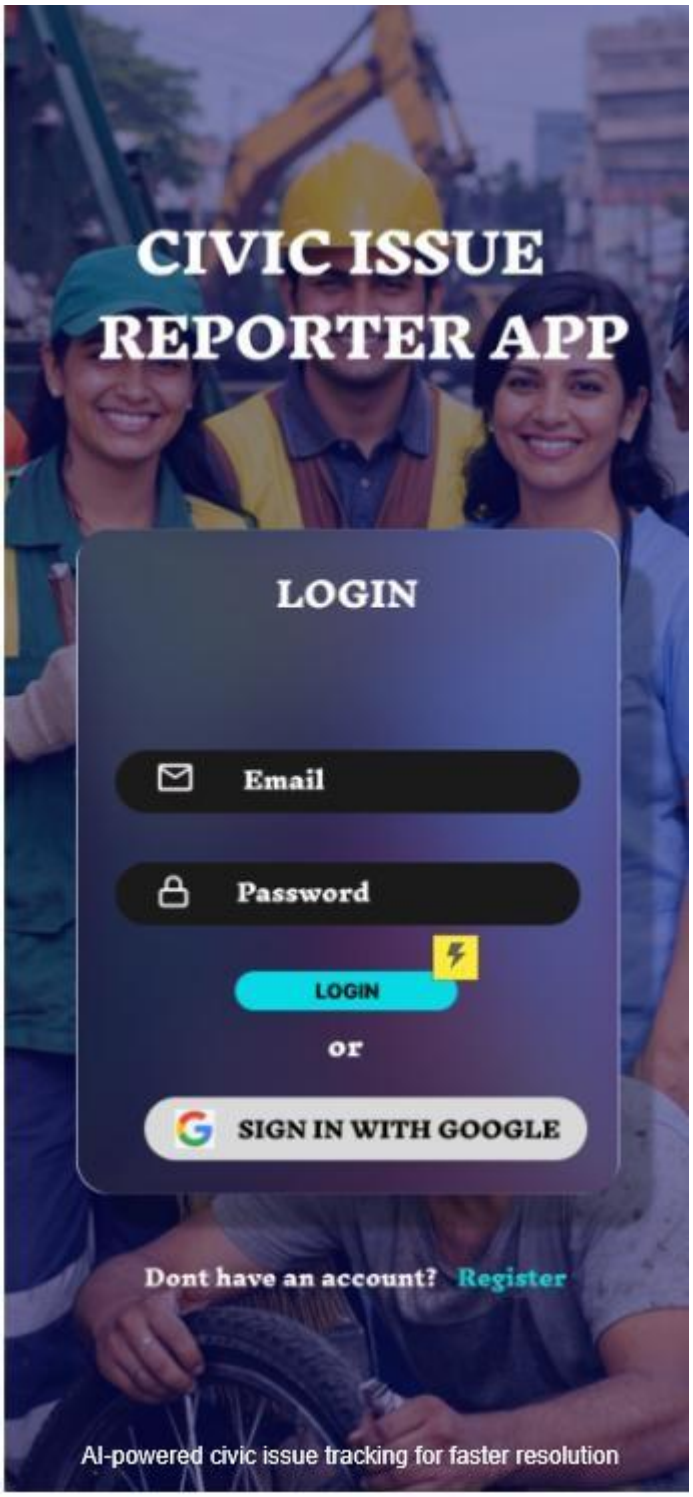
PROCEDURE:

Step 1: Designing Login Interface

Description:
A login interface is created to allow users to securely access the application. It includes email, password fields, and Google sign-in option.

RAG Concept Applied:
No RAG is applied here as authentication does not require data retrieval or AI processing.

Output:



Step 2: Designing Dashboard (Home Screen)

Description:
A dashboard is created displaying complaint status and navigation options for reporting issues.

RAG Enhancement:
A smart message is added:

“AI will prioritize your reports for faster action”

This simulates intelligent system behavior.

Output:

Step 3: Designing Issue Reporting Form

Description:
A form is designed where users can upload images, enter descriptions, and submit issues.

RAG Enhancement:
A smart hint is added:

“AI will analyze your issue and suggest the right department”

This represents classification using retrieved data.

Output:

Step 4: Designing Issue Preview Screen

Description:
Before submission, the user can review the issue details including image and description.

RAG Enhancement:
A classification message is added:

"AI detected: Likely category — Sanitation"

This simulates retrieval and matching with existing issue categories.

Output:

Step 5: Designing Complaint Tracking Screen

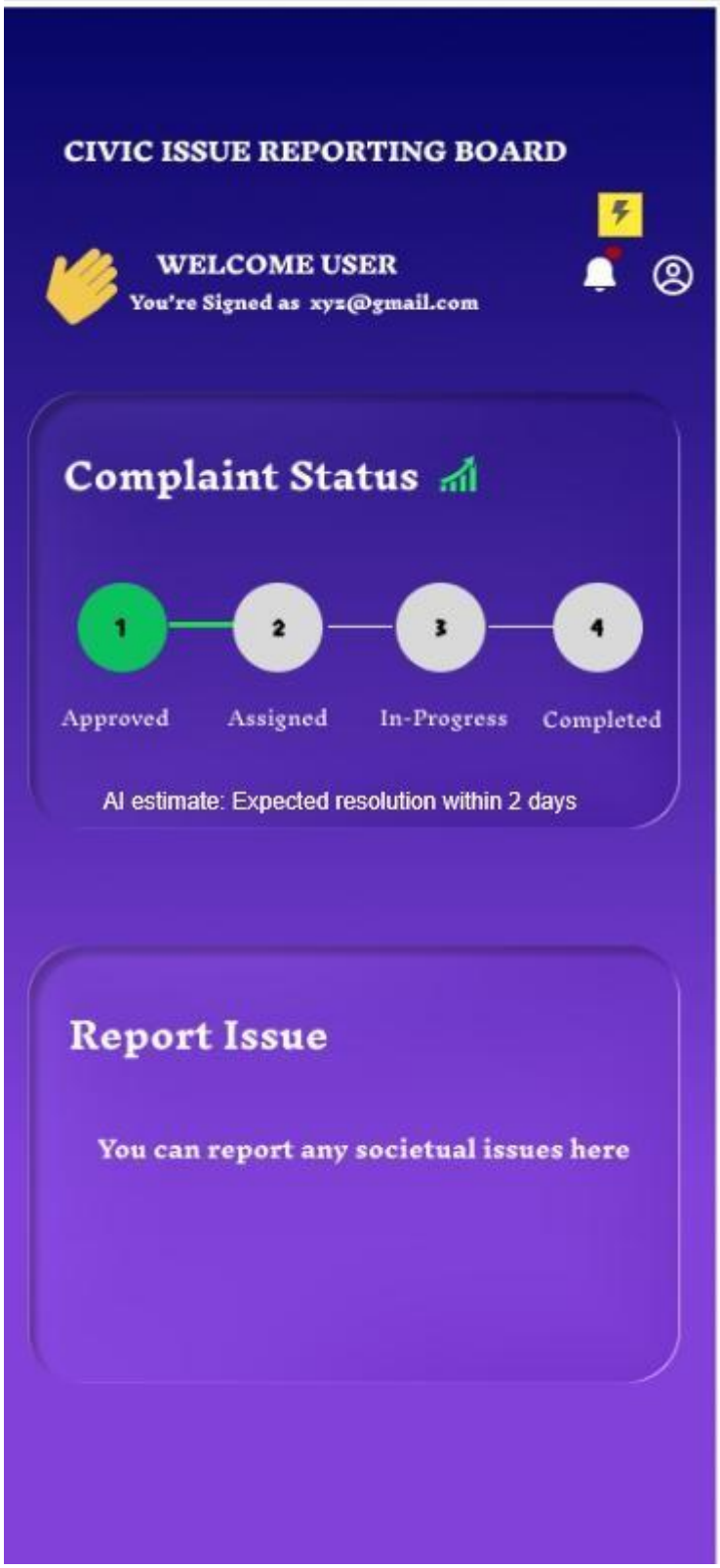
Description:
A status tracking interface is created showing stages like Approved, Assigned, In Progress, and Completed.

RAG Enhancement:
A prediction message is added:

"AI estimate: Expected resolution within 2 days"

This represents prediction using past data.

Output:


CIVIC ISSUE REPORTING BOARD
WELCOME USER
You're Signed as xyz@gmail.com
Complaint Status
1
2
3
4
Approved
Assigned
In-Progress
Completed
AI estimate: Expected resolution within 2 days
Report Issue
You can report any societual issues here

Step 6: Designing Notification System

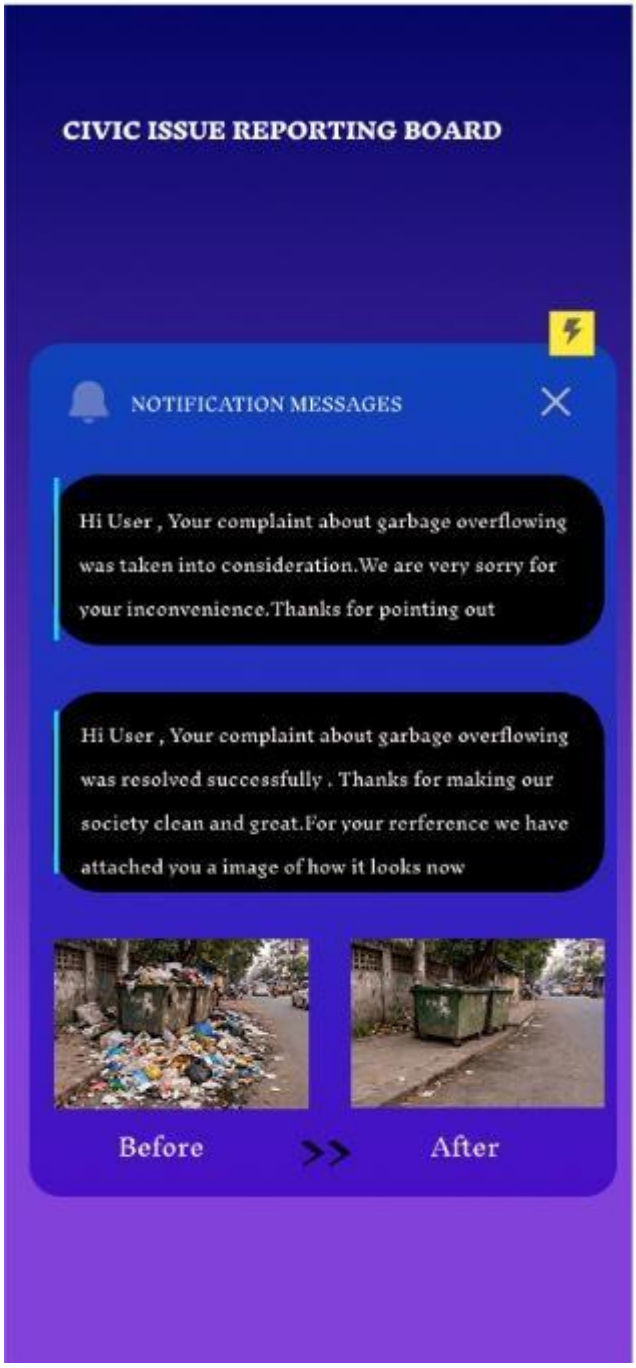
Description:
A notification panel is created to inform users about complaint updates.

RAG Enhancement:
A contextual message is added:

"AI matched this issue with similar cases in your area"

This simulates retrieval of similar complaints.

Output:



RESULT:

The Civic Issue Reporting System was successfully designed with RAG-inspired features. The application provides intelligent feedback such as issue classification, prediction of resolution time, and contextual notifications, improving user interaction.

CONCLUSION:

This experiment demonstrates how RAG concepts can be effectively simulated using UI design. Even without complex backend implementation, the system provides a smart and interactive experience, showcasing the potential of AI in civic applications.